

Demande:

$$\int_{-\infty}^{+\infty} \text{sinc}\left(\frac{t}{T}\right) \text{sinc}\left(\frac{0.1t}{T}\right) dt = \int_{-\infty}^{+\infty} \frac{T \cdot T}{0.1} \text{rect}\left(\frac{f}{T}\right) \text{rect}\left(\frac{f}{0.1}\right) df$$

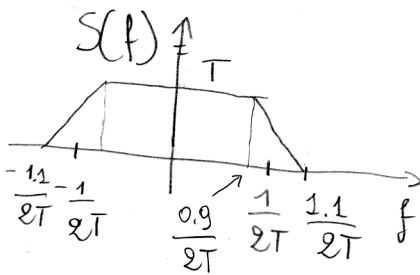
$$= \int_{-\infty}^{+\infty} \frac{T \cdot T}{0.1} \text{rect}\left(\frac{f}{T}\right) df = T$$

$$\left[\text{area} \left\{ \text{sinc}\left(\frac{t}{T}\right) \text{sinc}\left(\frac{0.1t}{T}\right) \right\} = T \right]$$

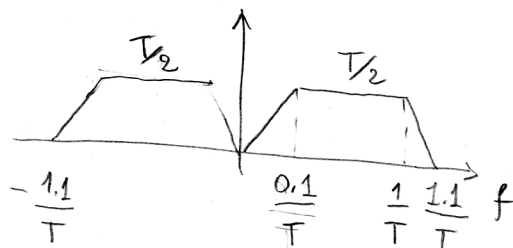
E1

$$x(t) = \text{sinc}\left(\frac{t}{T}\right) \text{sinc}\left(\frac{0.1t}{T}\right) \cos 2\pi \frac{1.1}{2T} t$$

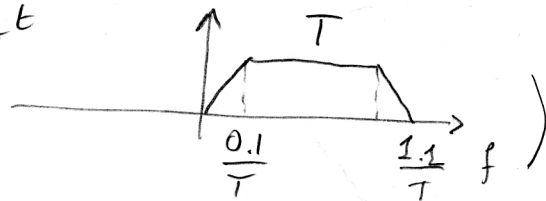
$$y(t) = \text{sinc}\left(\frac{t}{T}\right) \text{sinc}\left(\frac{0.1t}{T}\right)$$



$$X(f) = \frac{1}{2} S\left(f - \frac{1.1}{2T}\right) + \frac{1}{2} S\left(f + \frac{1.1}{2T}\right)$$

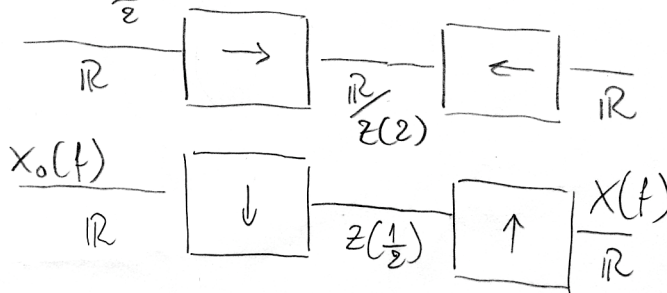


$$x(t) = \text{sinc}\left(\frac{t}{T}\right) \text{sinc}\left(\frac{0.1t}{T}\right) e^{j\pi \frac{1.1}{2T} t}$$



E2

$$x_d(t) = t \text{ rect}\left(\frac{t}{2}\right)$$



$$X(f) = \frac{1}{2} \sum_{k=-\infty}^{+\infty} X_0\left(\frac{k}{2}\right) \delta_{\mathbb{R}}\left(f - \frac{k}{2}\right)$$

$$\begin{aligned}
 X_0(f) &= \int_{-\infty}^{+\infty} t \operatorname{rect} \frac{t}{2} e^{-j2\pi f t} dt = \int_{-1}^1 t e^{-j2\pi f t} dt \\
 &= \frac{1}{-j2\pi f} t e^{-j2\pi f t} \Big|_{-1}^1 - \frac{1}{-j2\pi f} \int_{-1}^1 e^{-j2\pi f t} dt \\
 &= j \frac{1}{\pi f} \cos 2\pi f - j \frac{1}{2\pi^2 f^2} \sin 2\pi f
 \end{aligned}$$

E3

$$\text{NOTA: } \cos^2 \alpha = \frac{1 + \cos 2\alpha}{2} \quad \sin^2 \alpha = \frac{1 - \cos 2\alpha}{2}$$

$$x(kT) = \cos^2 \frac{k\pi}{2} = \frac{1}{2} + \frac{1}{2} \cos k\pi = \frac{1}{2} + \frac{1}{2} e^{j kT 2\pi \frac{1}{2T}}$$

$$X(f) = \frac{1}{2} \delta_{\frac{1}{2T}}\left(\frac{f}{\frac{1}{2T}}\right) + \frac{1}{2} \delta_{\frac{1}{2T}}\left(\frac{f - \frac{1}{2T}}{\frac{1}{2T}}\right)$$

E4

$$H_z(z) = T \frac{z - 1.3z^{-1}}{1 - 1.3z^{-1} + 0.4z^{-2}} = \frac{T}{1 - 0.8z^{-1}} + \frac{T}{1 - 0.5z^{-1}}$$

$$P_{1,2} = \frac{1.3 \pm \sqrt{(1.3)^2 - 1.6}}{2} = \begin{matrix} 0.8 \\ 0.5 \end{matrix}$$

$$h(kT) = (0.8)^k 1_0(kT) + (0.5)^k 1_0(kT)$$